

Effects of Different Demographics when Evaluating Interactive Robot-Empowered Language Learning in Human-Robot Interaction

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Abstract—The previous work developed a language learning system with human-like robots utilizing self-determination theory (SDT) to enhance student' learning engagement and proficiency. Initial evaluations indicated that typical students encountered stimulated interest and efficacy with the robot. To test the consistency of these findings, we conducted a replication study using the same system, robot, and tasks but with students with dyslexia. Through questionnaires, observations, and proficiency tests, the results confirmed that the interactive robot also improved learning motivation and performance in those with dyslexia, with the same system, robot, and tasks.

Index Terms—Socially Interactive Robot, Self-determination Theory, Human-robot Interaction (HRI), Dyslexia, Engagement

I. INTRODUCTION

Most studies in human-robot interaction (HRI) that introduce new advancements in robot interaction, behaviour, emotion, cognition, motivation, and learning efficacy often overlook testing their technology in diverse scenarios, user groups, or contexts, with some exceptions. The limitation impacts the validity of generalizing results for each technological change, as these variations significantly influence how users perceive the robots and their overall interaction experience.

Fung *et al.* [1] contributed to HRI research on learning motivation and performance by proposing a human-like robot-assisted language learning system based on self-determination theory (SDT), demonstrated in an empirical study with typical students learning Traditional Chinese and Cantonese. They

evaluated the effectiveness of their system by performing the empirical study with typical students in a classroom setting that explored two research questions (RQs): RQ1) To what extent can human-like robots impact students' motivation and engagement in language learning? RQ2) How do human-like robots affect students' proficiency in language learning?

Prior research has demonstrated that incorporating human-like functionalities into educational systems can foster student participation [2], [3]. These functionalities can enable dynamic interactions, promote positive learning behaviours, and support active involvement. The treatment group (learning with robots) improved engagement and motivation. Simultaneously, the control group (learning with teachers) slightly improved cognitive engagement but decreased emotional and behavioural engagement and motivation. Also, the treatment group showed a considerable improvement in learning scores.

In this study, we attempted to replicate a previous evaluation using the same methodology, including the robot, theoretical framework, pedagogy, game interface, system, interaction design, motivation strategies, and similar research questions. We examined the reproducibility of prior findings in a new context: students with dyslexia interacting with the robot in a school setting. We intended to adjust the participant population to expand the generalizability of results from typical students to those with dyslexia. We anticipated that this demographic would be more engaged with the tasks and that repeating the previous study with students with dyslexia would yield more apparent results in addressing our research questions.

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II. RELATED WORK

Active learning is challenging for students, particularly those with dyslexia, who experience difficulties in reading and writing [4]. They are more susceptible to demotivation when confronted with repetitive or monotonous tasks [5]. Educational technology, especially interactive tools such as robots, has demonstrated the potential to enhance students' learning interests. Recent research has spotlighted the effectiveness of assistive technology in improving students' learning experience and motivation [6].

A. Robot-empowered Learning

Robots have been extensively utilized in education, particularly in primary school settings. Wu *et al.* [7] introduced a system incorporating object recognition technology with a robot to teach English to Chinese children. The system enhanced language learning efficiency through interactive elements, such as touch and movement. Similarly, Fung *et al.* [8] employed human-like robots to boost students' learning proficiency, leveraging multimedia components and instant feedback to increase student interest in learning. Additionally, Andruseac *et al.* [9] created a training program that utilized robotics to teach science and technology to children with dyslexia. Their findings indicated that robots could assist students in developing problem-solving skills and learning computer programming, mathematics, and science.

B. Self-determination Theory (SDT)

By emphasizing digital autonomy, competence, and relatedness, SDT promotes the intrinsic motivation and learning engagement [10]. Robots can potentially advocate language learning through customized feedback, engaging interactive tasks, and social exchanges, thereby elevating interest in learning outcomes. Behavioural engagement is characterized by active involvement and sustained effort in learning endeavours [11]. This approach can be tailored to conform to social norms and language rules, thereby improving children's abilities in language and communication. Emotional engagement captures how learners respond emotionally and invest themselves in their tasks [12]. Robots can strengthen emotional connection by fostering a supportive atmosphere and employing affective computing to gauge learners' emotional states. Cognitive engagement means the involvement in the active processing of information in deep learning [13]. Robots' physical presence can intensify cognitive engagement by lending them perceived authority. Intrinsic motivation drives exploration and curiosity, encouraging activity engagement [14]. Research shows that social robots can stimulate intrinsic motivation, with findings indicating improved task engagement strength and duration.

C. Interactive Elements

Recent studies have explored various interactive design features to impact learning curiosity [15]–[17]. Sun and Hsieh [18] combined gamification and interactive elements in classroom activities. They found that the fun and interactive nature can improve students' learning interests and attention.

Similarly, Neumann [19] and Barnes *et al.* [20] observed that children are more eager to learn with educational interactions when robots incorporate elements such as singing and dancing.

Interactive features (i.e., gesture, mobility, and animacy) can boost students' involvement in learning. Gesture refers to expressions used by human-like robots to communicate with users [21]. They are important for human cognition and significantly drive students' active participation [22]. Incorporating gestures in robot-empowered learning fosters effective involvement [23], with specific gestures like shaking hands and nodding head as positive reinforcements [24]. Mobility is another critical interactive design feature that can capture students' attention and stimulate curiosity [25]. For example, Robins *et al.* [26] found that the physical movement of human-like robots significantly increased five key developmental areas: sensory, communication and interaction, motor, cognitive, and social and emotional. Animacy describes how similar a robot mirrors human-like characteristics, which has been shown to improve learning experiences. Ahmad *et al.* [27] revealed that animation can sustain children's engagement. Also, human-like robots that effectively convey emotions foster a more empathetic atmosphere conducive to learning [28].

III. SYSTEM DESIGN

We utilized multiple functionalities and humanoid robots to align the concepts of autonomy, competence, and relatedness in SDT. To drive a sense of ownership over the learning journey, the system provided students with customizable features, such as re-listening to questions, detecting threshold, and personalized speech rate selection, to empower students to tailor their learning experience. Students also had three trials to answer the questions, and the robot would teach them or reinforce the correct answer for memory consolidation.

The system was programmed to offer interactive feedback, allowing students to navigate their learning path actively. To strive for competence, the system provided students with a structured progression, from vocabulary to sentence expansion. The scaffolding method allows students to build on their existing knowledge. The system also implemented social interactions to boost relatedness. For example, the robot can mimic human-like behaviours and provide a friendly interaction (e.g., cheering and thumb up) to make students feel more comfortable and connected, enhancing their willingness to participate. The system also provided students with four learning sets, each consisting of 13 games that covered vocabulary (3), short sentence (5), and sentence expansion (5). Students had a maximum of three chances to answer each question, with 10 seconds for recording or pressing to stop. No matter students answered the questions correctly, incorrectly, or with no response, the system was able to respond to students with relevant facial expressions, supportive gestures, and encouraging verbal support. After finishing a learning set, the robot sang a song to praise the students.

IV. EXPERIMENT

This section summarizes the experiment to evaluate the effectiveness of interactive robot and addressed the RQs. This study employed a triangulation approach, using questionnaires, observations, and proficiency tests to measure student interest. Two groups of students were assigned by school teachers based on the inclusion criteria, ensuring they had comparable demographics: a control and treatment group. The learning materials of both groups were in the same difficulty level.

a) *Inclusion Criteria:* Regarding the inclusion criteria for participation, students studied in Grades 2 – 5. They were able to read Traditional Chinese and speak Cantonese. Also, they did not have any medical or physical disabilities that might hinder the interaction with robots and communication. Moreover, they had the experience to use digital tools. Students participated in the study voluntarily and without any remuneration. Parental consent was received from the guardians prior to running the experiment. The University Institutional Review Board (IRB) approved the experiment protocol.

b) *Questionnaires:* The pre-/post questionnaires [1] measured four factors related to need fulfillment and student involvement on a 5-point Likert scale. An experienced special education coordinator (SENCO) reviewed the items to ensure they were accessible for those with dyslexia. Students completed the questionnaire before and after the learning period. By maintaining the rigour of the methodology and ensuring a similar structured learning environment, we aimed to clearly understand the differences in engagement and motivation between the two conditions. This approach provided us with insights regarding the benefits of robot-assisted learning for elevating student outcomes compared to conventional teacher-led instruction.

c) *The Treatment Group:* We invited 90 students (32 females and 58 males) from 9-year-old to 12-year-old ($\bar{M}$ = 10.58-year-old, σ = 0.81) from five Hong Kong primary schools with similar level. Students completed the questionnaires on Days 1 and 5 and engaged in the robot training game over 5 days, with one set per day. From Days 1 to 4, students progressed through Sets 1 to 4, with each set increasing in difficulty. On the fifth day, they completed Set 1 again (post-test) to assess their learning efficacy through language proficiency tests. Each set of tests took approximately 20 minutes to complete. For the training sessions in the classroom, six students were arranged in one classroom, each student was assigned a table, which was positioned two meters away from adjacent tables to reduce distractions (e.g., social comparison).

d) *The Control Group:* We invited 40 students (14 females and 26 males) from 9-year-old to 13-year-old ($\bar{M}$ = 11.52-year-old, σ = 1.26) from the five primary schools to join our study. Students filled out the questionnaires on Days 1 & 5. Six students participated in a conventional classroom setting led by an experienced Chinese teacher. The teacher was responsible for preparing and designing the learning materials aligned with the content and complexity of the materials used in the experimental group. This alignment is to ensure a fair comparison of engagement and motivation performance.

V. DATA ANALYSIS

In the following section, we analyzed the collected data to answer two research questions. **RQ1** examined how interactive elements impacted students' learning engagement and motivation. In general, the performance between the two studies was similar. As depicted in Fig. 1, the ANCOVA results indicated that learning with robots significantly drove students' active participation compared to conventional learning. In contrast to typical students, those with dyslexia in the control group exhibited a greater drop in intrinsic motivation. The detailed analysis is shown as follows.

The treatment group demonstrated a considerable improvement in *behavioural engagement*, whereas the control group encountered a modest drop. The treatment group did not exhibit statistical significance (pre-test: $\bar{M}$ = 11.24, σ = 2.36, post-test: $\bar{M}$ = 11.92, σ = 2.94, p = .09, f - value = 2.91). The control group also did not exhibit statistical significance (pre-test: $\bar{M}$ = 11.10, σ = 2.54, post-test: $\bar{M}$ = 11.90, σ = 2.59, p = .17, f - value = 1.95).

The treatment group manifested a sounded improvement in *emotional engagement*. However, the control group experienced a minor decrease. The treatment group exhibited statistical significance (pre-test: $\bar{M}$ = 10.51, σ = 2.85, post-test: $\bar{M}$ = 12.64, σ = 2.75, p < .001, f - value = 26.13). The control group also did not exhibit statistical significance (pre-test: $\bar{M}$ = 10.78, σ = 2.83, post-test: $\bar{M}$ = 11.00, σ = 2.76, p = .72, f - value = .13).

The treatment group illustrated a remarkable improvement in *cognitive engagement*, while the control group had a slight increase. The treatment group exhibited statistical significance (pre-test: $\bar{M}$ = 10.57, σ = 2.57, post-test: $\bar{M}$ = 12.11, σ = 2.88, p < .001, f - value = 14.40). The control group also did not exhibit statistical significance (pre-test: $\bar{M}$ = 11.23, σ = 2.59, post-test: $\bar{M}$ = 10.90, σ = 2.62, p < .58, f - value = .13).

The treatment group expressed a substantial progress in *intrinsic motivation*, but the control group experienced a small drop. The treatment group exhibited statistical significance (pre-test: $\bar{M}$ = 10.83, σ = 3.22, post-test: $\bar{M}$ = 11.81, σ = 3.28, p < .05, f - value = 4.07). The control group also did not exhibit statistical significance (pre-test: $\bar{M}$ = 10.50, σ = 2.92, post-test: $\bar{M}$ = 10.13, σ = 3.20, p = .59, f - value = .30).

The evaluation shown that the control group exhibited a bit more engagement compared to the treatment group at the beginning. After five days' learning with robots, the treatment group had a promising enhancement. However, the control group showed a decrease in participation and enthusiasm. Also, the variation within the treatment group was less disperse. These findings were more pronounced than those in the previous study, suggesting that interactive robot-empowered learning positively contributed to active engagement for those with dyslexia in technology-based learning environments.

RQ2 explored how interactive robots facilitated learning efficacy in the treatment group. As such, we utilized single-sample paired t-tests to analyze students' learning outcomes

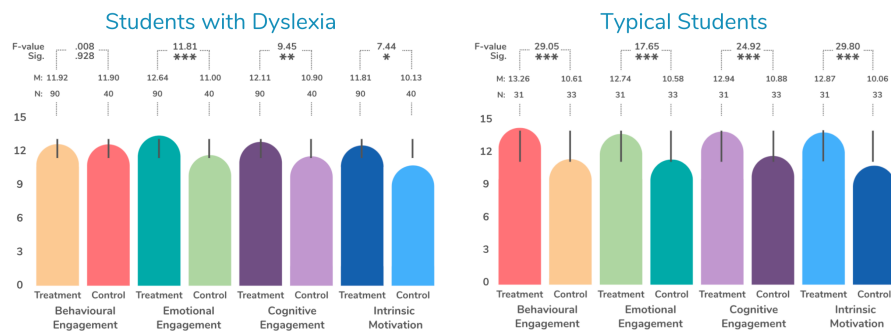


Fig. 1. The figures depicted the overview of analysis of covariance (ANCOVA) with error bars of standard deviation for engagement and motivation in both experiments. ***, ** and * denote $p < .001$, $p < .005$ and $p < .01$, respectively.

TABLE I
OVERVIEW OF SINGLE-SAMPLE PAIRED T-TESTS FOR BOTH EXPERIMENTS.
THE ACRONYMS IMP. DENOTES IMPROVEMENT.

		<i>M</i>	σ	N	df	Imp.	Sig.
Dyslexia	Pre	0.90	11.95	90	89	5.03	< .005
	Post	0.95	7.67	90	89		
Typical	Pre	0.81	0.15	31	30	9.88	< .001
	Post	0.89	0.13	31			

(Table I). The results indicated that the treatment group achieved a substantial enhancement of 17.26% ($p < .001$). It was a more sound improvement than the prior study by 9.88% ($p < .05$). These results confirmed that all prerequisites were satisfied for executing the paired t-tests for RQ2. During the five-day training, the interactive robot markedly improved learning performance in the treatment group. The results supported by SDT, suggesting that when psychological needs are effectively addressed through pedagogical design, students with dyslexia are more active in learning independently.

VI. DISCUSSION

The replication study results reported in this work suggest that students with dyslexia supported the RQs in the prior work. In this section, we discuss the generalizability of findings and explore how interactive robot-empowered learning can impact a demographic that often faces unique challenges in educational settings.

a) Engagement as a Catalyst for Learning: Significant improvements in behaviour, emotion, and cognition in students with dyslexia underscore engagement's crucial role in learning. Engaged students show persistence, resilience, and better outcomes. The engaging features of human-like robots (i.e., personalized feedback and adaptive learning) foster a stimulating and supportive environment, emphasizing that engagement is not just a precursor but integral to learning [29]–[31].

b) The Role of Autonomy and Competence: The findings highlight the significance of autonomy and competence. Individualized learning raise students' inherent motivation, while traditional teacher-led instruction decreases it. This suggests that a lack of autonomy may hinder learning intention [32]. Personalized learning is essential for empowering students, especially those with dyslexia, to take ownership of their study.

c) Learning Proficiency Enhancements: The treatment group showed a significant improvement in learning scores, with a 17.26% increase in proficiency, surpassing the previous study's 9.88% Improvement in typical students. This suggests that the interactive robot enhances engagement and leads to better learning outcomes [33], making a strong case for using robotic technology in education for students with dyslexia.

d) The Performance of the Control Group: The control group had a minimal progress in emotional and behavioural engagement but a deterioration in cognitive engagement and motivation. The performance may lead to lower cognitive engagement, as students may be affected by receiving information passively or being influenced by peers.

e) Implications for Educational Practice: These findings have important implications for educational practice. Human-like robots can be especially helpful in classrooms for students with dyslexia, as a supportive learning environment can encourage active participation [34]. By addressing the unique needs of those with dyslexia, educators can create a more inclusive and effective learning atmosphere.

f) Limitations: While this study contributes to understanding robot-empowered learning, it is essential to acknowledge its limitations. The sample size was small, and the study was conducted in a single educational context.

g) Future Research: Future research can explore the long-term effects of robot-empowered learning on different types of special learning disability (e.g., dyslexia, ADHD and autism), across various settings with larger and more diverse samples. Also, investigating the specific features of robots that most beneficial to learning engagement and outcomes could provide deeper insights into optimizing their educational use.

VII. CONCLUSION

In conclusion, this study demonstrates that human-like robots can significantly enhance learning engagement and proficiency among students with dyslexia. By leveraging the principles of SDT and focusing on interactive elements, educators can create more engaging learning environments to cater learners' unique needs. This research paves the way for further exploration of the role of technology in education, particularly in supporting students with learning difficulties.

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